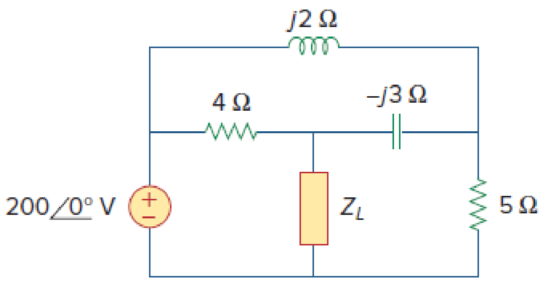
Chapter 11, Problem 12P

Problem

For the circuit shown in Fig. 11.44, determine the load impedance ***ZL*** for maximum power transfer (to ***ZL***). Calculate the maximum power absorbed by the load.

Figure 11.44:



Step-by-step solution

Step 1 of 13

Refer to circuit diagram in Figure 11.44 in the text book.

Deactivate the sources and redraw the circuit diagram as shown in Figure 1 to calculate the Thevenin's impedance.

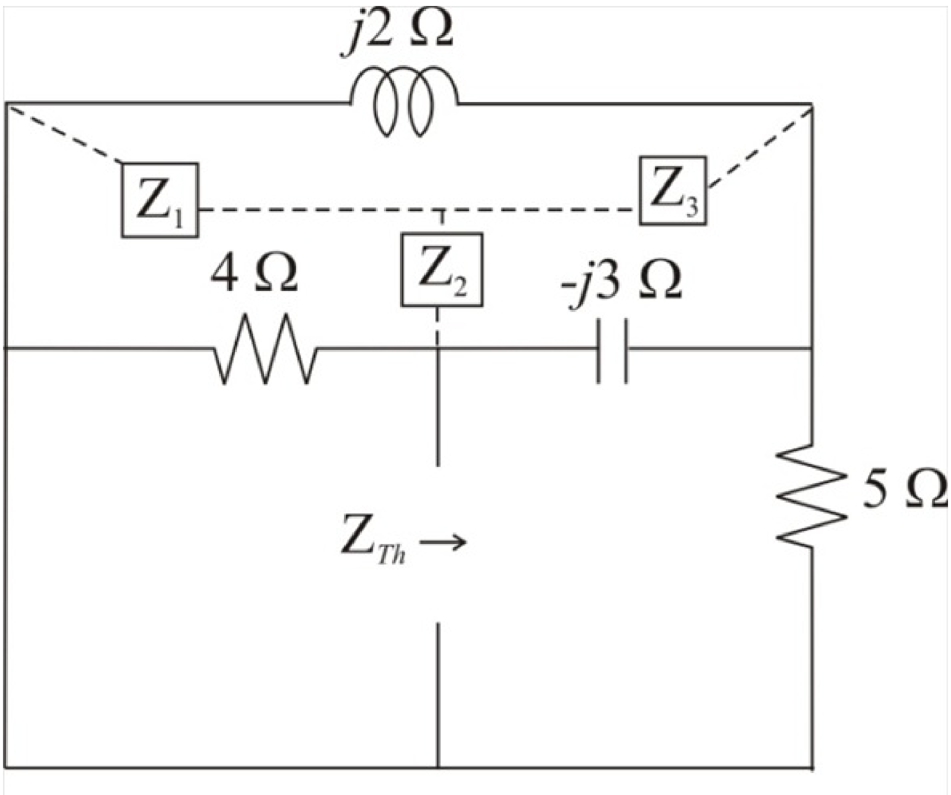


Figure 1

Step 2 of 13

Convert the delta connected to wye connection.

Calculate the value of the impedance Z_1.

$$\begin{aligned} Z_1 &= \frac{(4)(j2)}{4+j2-j3} \\ &= \frac{j8}{4-j1} \\ &= -0.471+j1.882\ \Omega \end{aligned}$$

Therefore, the value of the impedance Z_1 is $-0.471+j1.882\ \Omega$.

Step 3 of 13

Calculate the value of the impedance Z_2.

$$\begin{aligned} Z_2 &= \frac{(4)(-j3)}{4+j2-j3} \\ &= \frac{-j12}{4-j1} \\ &= 0.706-j2.823\ \Omega \end{aligned}$$

Therefore, the value of the impedance Z_2 is $0.706-j2.823\ \Omega$.

Step 4 of 13

Calculate the value of the impedance Z_3.

$$\begin{aligned} Z_3 &= \frac{(j2)(-j3)}{4+j2-j3} \\ &= \frac{6}{4-j1} \\ &= 1.412+j0.353\ \Omega \end{aligned}$$

Therefore, the value of the impedance Z_3 is $1.412+j0.353\ \Omega$.

Step 5 of 13

Redraw the circuit diagram as shown in Figure 1.

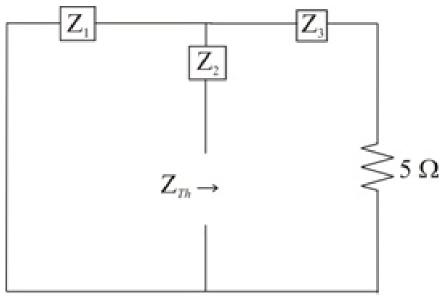


Figure 2

Step 6 of 13

Calculate the value of the Thevenin's impedance Z_{Th}.

$$Z_{Th} = Z_2 + \left[Z_1 \| (Z_3 + 5)\right]$$

Substitute $1.412 + j0.353\ \Omega$ for Z_3, $0.706 - j2.823\ \Omega$ for Z_2 and $-0.471 + j1.882\ \Omega$ for Z_1.

$$\begin{aligned} Z_{Th} &= 0.706 - j2.823 + \left[(-0.471 + j1.882) \| (1.412 + j0.353 + 5)\right] \\ &= 0.706 - j2.823 + \left[(-0.471 + j1.882) \| (6.412 + j0.353)\right] \\ &= 0.706 - j2.823 + \frac{(-0.471 + j1.882)(6.412 + j0.353)}{(-0.471 + j1.882) + (6.412 + j0.353)} \\ &= 0.706 - j2.823 + 0.117 + j1.96 \end{aligned}$$

$$= 0.823 - j0.864\ \Omega$$

Therefore, the value of the Thevenin's impedance Z_{Th} is $0.823 - j0.864\ \Omega$.

The value of the Thevenin's resistance R_{Th} is $0.823\ \Omega$.

Step 7 of 13

Calculate the value of the load impedance Z_L.

$$Z_L = Z_{Th}^*$$

Substitute $0.823 - j0.864\ \Omega$ for Z_{Th}.

$$\begin{aligned} Z_L &= (0.823 - j0.864)^* \\ &= 0.823 + j0.864\ \Omega \end{aligned}$$

Therefore, the required value of the load impedance Z_L is $\boxed{0.823 + j0.864\ \Omega}$.

Step 8 of 13

Consider the circuit diagram shown in Figure 1 to calculate the value of the Thevenin's voltage.

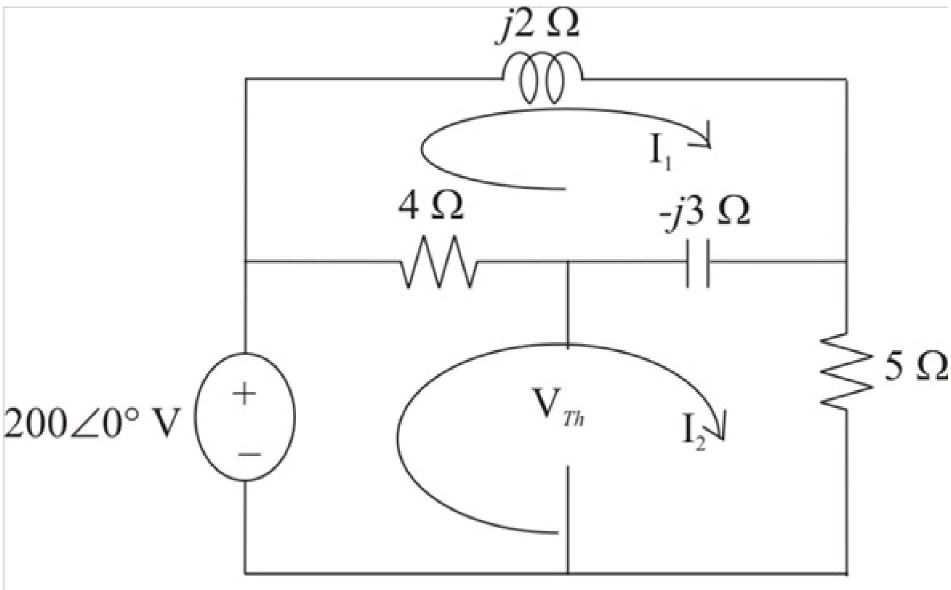


Figure 2

Step 9 of 13

Apply Kirchhoff's voltage law in Loop-1.

$$\mathbf{I}_1(j2)+(\mathbf{I}_1-\mathbf{I}_2)(4-j3)=0$$
$$\mathbf{I}_1(j2+4-j3)-\mathbf{I}_2(4-j3)=0$$
$$\mathbf{I}_1(4-j1)=\mathbf{I}_2(4-j3)$$
$$\mathbf{I}_1=\left(\frac{4-j3}{4-j1}\right)\mathbf{I}_2 \;\ldots\ldots\; (1)$$

Therefore, the expression for the current I_1 is $\mathrm{I}_2\left(\frac{4-j3}{4-j1}\right)$.

Step 10 of 13

Apply Kirchhoff's voltage law in Loop-2.

$$-200\angle 0°+(\mathbf{I}_2-\mathbf{I}_1)(4-j3)+\mathbf{I}_2(5)=0$$
$$(\mathbf{I}_2)(4-j3+5)-\mathbf{I}_1(4-j3)=200\angle 0°$$
$$(\mathbf{I}_2)(9-j3)-\mathbf{I}_1(4-j3)=200\angle 0°$$

Substitute $\left(\frac{4-j3}{4-j1}\right)\mathbf{I}_2$ for $\mathbf{I}_1$.

$$(9-j3)\mathbf{I}_2-(4-j3)\left(\frac{4-j3}{4-j1}\right)\mathbf{I}_2=200\angle 0°$$
$$(9-j3)\mathbf{I}_2-(3.058-j5.235)\mathbf{I}_2=200\angle 0°$$
$$(5.94+j2.235)\mathbf{I}_2=200\angle 0°$$
$$\begin{aligned}\mathbf{I}_2&=\frac{200\angle 0°}{5.94+j2.235}\\&=31.51\angle -20.62°\ \mathrm{A}\end{aligned}$$

Therefore, the value of the current $\mathbf{I}_2$ is $31.51\angle -20.62°\ \mathrm{A}$.

Step 11 of 13

Recall equation (1).

$$\mathbf{I}_1=\left(\frac{4-j3}{4-j1}\right)\mathbf{I}_2$$

Substitute $31.51\angle -20.62°\ \mathrm{A}$ for $\mathbf{I}_2$.

$$\begin{aligned}\mathbf{I}_1&=\left(\frac{4-j3}{4-j1}\right)(31.51\angle -20.62°\ \mathrm{A})\\&=(1.213\angle -22.83°)(31.51\angle -20.62°\ \mathrm{A})\\&=38.21\angle -43.45°\ \mathrm{A}\end{aligned}$$

Therefore, the value of the current I_1 is $38.22\angle -43.45°\ \mathrm{A}$.

Step 12 of 13

Calculate the value of the Thevenin's voltage.

$$\mathbf{V}_{\text{Th}} = 200\angle 0° - 4(\mathbf{I}_2 - \mathbf{I}_1)$$

Substitute $38.21\angle -43.45° \text{ A}$ for $\mathbf{I}_1$ and $31.51\angle 20.62° \text{ A}$ for $\mathbf{I}_2$.

$$\begin{aligned}\mathbf{V}_{\text{Th}} &= 200\angle 0° - (4)(31.51\angle 20.62° - 38.21\angle -43.45°)\\ &= 200\angle 0° - (4)(15.28\angle 83.42°)\\ &= 200\angle 0° - 61.13\angle 83.42°\\ &= 202.32\angle -17.46° \text{ V}\end{aligned}$$

Therefore, the value of the Thevenin's volage $\mathbf{V}_{\text{Th}}$ is $202.32\angle -17.46° \text{ V}$.

Step 13 of 13

Calculate the value of the maximum power absorbed by the load.

$$P_{\max} = \frac{|\mathbf{V}_{\text{Th}}|^2}{8R_{\text{Th}}}$$

Substitute $0.823\ \Omega$ for R_{Th} and 202.32 V for $|\mathbf{V}_{\text{Th}}|$.

$$\begin{aligned}P_{\max} &= \frac{(202.32 \text{ V})^2}{8(0.823\ \Omega)}\\ &= \frac{40933.38}{6.584} \text{ W}\\ &= 6.217 \text{ kW}\end{aligned}$$

Therefore, the value of the maximum power $P_{\max}$ absorbed by load is $\boxed{6.217 \text{ kW}}$.